

IMO Training – Number Theory

IMO Shortlist Problems

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Notes

Unless otherwise specified, let p be a prime number.

Theorem. (Fermat's Little Theorem) $\forall a \in \mathbb{Z}, a^p \equiv a \pmod{p}$. Equivalently, $\forall a \in \mathbb{Z}$ with $(a, p) = 1$, $a^{p-1} \equiv 1 \pmod{p}$.

Definition. (Euler Totient Function) $\forall n \in \mathbb{N}$, $\varphi(n) = \#\{m \in \mathbb{N} \mid 1 \leq m < n, (m, n) = 1\}$. Equivalently, for $n = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}$, where $p_1, p_2, \dots, p_k$ being distinct primes, and integers $a_1, a_2, \dots, a_k \geq 1$, $\varphi(n) = n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \dots \left(1 - \frac{1}{p_k}\right)$.

Theorem. (Fermat-Euler Theorem) $\forall a \in \mathbb{Z}, n \in \mathbb{N}$ s.t. $(a, n) = 1$, $a^{\varphi(n)} \equiv 1 \pmod{n}$.

Theorem. (Wilson's Theorem) p is a prime iff $(p-1)! \equiv -1 \pmod{p}$.

Definition. (Order and Primitive Roots) For $a \in \mathbb{Z}, n \in \mathbb{N}$ with $(a, n) = 1$, the *order of a modulo n* ($\text{ord}_n(a)$) is defined by the minimal $m \in \mathbb{N}$ s.t. $a^m \equiv 1 \pmod{n}$. Furthermore, if $\text{ord}_n(a) = \varphi(n)$, then a is a *primitive root modulo n* .

Definition. (Valuation) For $m \in \mathbb{N}$, the *m -adic valuation* of n is the $a \in \mathbb{N}$ s.t. $m^a \mid n$ and $m^{a+1} \nmid n$.

Theorem. If p odd, $(x, p) = (y, p) = 1$ and $p \mid x - y$, then for all $n \in \mathbb{N}$

$$\nu_p(x^n - y^n) = \nu_p(x - y) + \nu_p(n).$$

Theorem. If $p, n > 0$ are odd, $(x, p) = (y, p) = 1$ and $p \mid x + y$, then

$$\nu_p(x^n + y^n) = \nu_p(x + y) + \nu_p(n).$$

Theorem. If $2 \mid x - y$, then for all even $n \in \mathbb{N}$

$$\nu_2(x^n - y^n) = \nu_2(x - y) + \nu_2(x + y) + \nu_2(n) - 1.$$

Theorem. If $4 \mid x - y$, then for all $n \in \mathbb{N}$

$$\nu_2(x^n - y^n) = \nu_2(x - y) + \nu_2(n).$$

1. (IMO 2010 N1)

Find the least $n \in \mathbb{N}$ for which $\exists$ a set $\{s_1, s_2, \dots, s_n\}$ consisting of n distinct positive integers, such that

$$\left(1 - \frac{1}{s_1}\right) \left(1 - \frac{1}{s_2}\right) \cdots \left(1 - \frac{1}{s_n}\right) = \frac{51}{2010}.$$

2. (IMO 2011 N1)

For any $d \in \mathbb{N}$, let $f(d)$ be the smallest possible integer that has exactly d positive divisors (e.g. $f(1) = 1, f(5) = 16, f(6) = 12$). Prove that for every $k \in \mathbb{Z}_{\geq 0}$, $f(2^k) \mid f(2^{k+1})$.

3. (IMO 2012 N2)

Find all triples (x, y, z) of positive integers such that $x \leq y \leq z$ and

$$x^3(y^3 + z^3) = 2012(xyz + 2).$$

4. (IMO 2013 N1)

Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$m^2 + f(n) \mid mf(m) + n.$$

5. (IMO 2013 N2)

Assume that k and n are two positive integers. Prove that there exist positive integers $m_1, \dots, m_k$ such that

$$1 + \frac{2^k - 1}{n} = \left(1 + \frac{1}{m_1}\right) \cdots \left(1 + \frac{1}{m_k}\right).$$

6. (IMO 2014 N2)

Find all pairs (x, y) of positive integers such that

$$\sqrt[3]{7x^2 - 13xy + 7y^2} = |x - y| + 1$$

7. (IMO 2014 N5)

Find all triples (p, x, y) consisting of a prime number p and two positive integers x and y such that $x^{p-1} + y$ and $x + y^{p-1}$ are both powers of p .

8. (IMO 2015 N1)

Find all $M \in \mathbb{N}$ such that the sequence $(a_n)_{n \geq 0}$ defined by

$$a_0 = M + \frac{1}{2} \text{ and } a_{k+1} = a_k \lfloor a_k \rfloor$$

contains at least one integer term.

9. (IMO 2015 N2)

Let a and b be positive integers such that $a! + b!$ divides $a!b!$. Prove that $3a \geq 2b + 2$.

10. (IMO 2015 N5)

Find all positive integers (a, b, c) such that $ab - c, bc - a, ca - b$ are all powers of 2.

11. (IMO 2016 N1)

For any $k \in \mathbb{N}$, denote the sum of digits of k in base 10 by $S(k)$. Find all polynomials $P(x)$ with integral coefficients such that for any positive integer $n \geq 2016$, $P(n) > 0$ and

$$S(P(n)) = P(S(n)).$$

12. (IMO 2016 N6)

Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that for all positive integers m and n , the integer $f(m) + f(n) - mn$ is nonzero and divides $mf(m) + nf(n)$.

13. (IMO 2017 N1)

For each integer $a_0 > 1$, define the sequence $(a_n)_{n \geq 0}$ by

$$a_{n+1} = \begin{cases} \sqrt{a_n} & \text{if } \sqrt{a_n} \text{ is an integer,} \\ a_n + 3 & \text{otherwise.} \end{cases}$$

Find all values of a_0 such that $\exists A$ with $a_n = A$ for infinitely many values of n .

14. (IMO 2017 N5)

Find all pairs (p, q) of prime numbers which $p > q$ and

$$\frac{(p+q)^{p+q}(p-q)^{p-q} - 1}{(p+q)^{p-q}(p-q)^{p+q} - 1}$$

is an integer.

15. (IMO 2018 N1)

Determine all pairs (a, b) of distinct positive integers such that $\exists k \in \mathbb{N}$ for which the number of divisors of ka and that of kb are equal.

16. (IMO 2019 N1)

Find all pairs (k, n) of positive integers such that

$$k! = (2^n - 1)(2^n - 2)(2^n - 4) \cdots (2^n - 2^{n-1}).$$

17. (IMO 2019 N2)

Find all $a, b, c \in \mathbb{N}$ such that

$$a^3 + b^3 + c^3 = (abc)^2.$$

18. (IMO 2019 N4)

Find all functions $f : \mathbb{Z}_{>0} \rightarrow \mathbb{Z}_{>0}$ such that $a + f(b)$ divides $a^2 + bf(a)$ for all positive integers a and b with $a + b > 2019$.

19. (IMO 2020 N1)

Given a positive integer k show that there exists a prime p such that one can choose distinct integers $a_1, a_2, \dots, a_{k+3} \in \{1, 2, \dots, p-1\}$ such that p divides $a_i a_{i+1} a_{i+2} a_{i+3} - i$ for all $i = 1, 2, \dots, k$.

20. (IMO 2020 N3)

A deck of $n > 1$ cards is given. A positive integer is written on each card. The deck has the property that the arithmetic mean of the numbers on each pair of cards is also the geometric mean of the numbers on some collection of one or more cards. For which n does it follow that the numbers on the cards are all equal?

21. (IMO 2022 Problem 5)

Find all triples (a, b, p) of positive integers with p prime and

$$a^p = b! + p$$

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